

Date of submission June 06, 2024.

Digital Object Identifier 077bct003/077bct033

Exploring Dimensionality Reduction with Principal Component Analysis (June 2024)

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ABSTRACT Principal Component Analysis (PCA) is a fundamental statistical technique extensively employed for data and dimensionality reduction across various domains. This paper delves into the mechanics and applications of PCA, elucidating its pivotal role in simplifying complex datasets, enhancing visualization, and refining predictive models. PCA aims to identify principal components that capture the maximum variance within high-dimensional data, subsequently projecting it onto a lower-dimensional space, with eigenvalues and eigenvectors aiding in the identification of the directions of maximum variance in the dataset. By mitigating the challenges associated with high-dimensional data, such as sparsity, computational complexity, overfitting, and distorted distance metrics, PCA facilitates efficient data analysis and interpretation. Leveraging synthetic and real-world datasets, including the 'Predict Students' Dropout and Academic Success' dataset from the UC Irvine Machine Learning Repository, this study provides practical insights into PCA's implementation. Rigorous preprocessing ensures data integrity, while Jupyter Notebook and libraries such as NumPy, Pandas, and Matplotlib facilitate seamless analysis and visualization. This comprehensive investigation underscores PCA's significance in empowering decision-making processes and advancing research across diverse domains, including education, social sciences, and machine learning.

INDEX TERMS Dimensionality reduction, Eigenvalues, Eigenvectors, Principal components, Real-world datasets, Statistical technique, Synthetic datasets, UC Irvine Machine Learning Repository

I. INTRODUCTION

Principal Component Analysis (PCA) is a powerful statistical technique used for data reduction and dimensionality reduction. It transforms high-dimensional data into a lower-dimensional form while retaining as much of the original variability as possible. This method is widely used in fields such as data science, machine learning, and bioinformatics, where it aids in simplifying complex datasets, enhancing visualization, and improving the performance of predictive models.

The core idea behind PCA is to identify the directions (principal components) along which the variance of the data is maximized. By projecting the data onto these principal components, PCA reduces the dimensionality of the data, which can help to reveal underlying patterns and structure that are not immediately apparent in the original high-dimensional space.

One of the key motivations for using PCA is to mitigate the effects of the "curse of dimensionality." This term refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not

occur in low-dimensional settings. As the dimensionality of the data increases, several issues can arise:

1. **Sparse Data:** In high-dimensional spaces, data points tend to be sparse. This sparsity can make it difficult to find meaningful patterns or relationships within the data because the points are so far apart.
2. **Computational Complexity:** The computational resources required for processing, storing, and analyzing high-dimensional data grow exponentially with the number of dimensions. This increase can lead to significant inefficiencies and longer processing times.
3. **Overfitting:** High-dimensional datasets often contain many variables, which can lead to models that fit the noise in the training data rather than the underlying signal. This overfitting results in poor generalization to new, unseen data.
4. **Distance Metrics:** In high dimensions, the notion of distance becomes less intuitive and less useful. For example, in high-dimensional spaces, the distance between the nearest and farthest points tends to converge,

making it difficult to distinguish between close and distant points.

PCA helps address these challenges by reducing the number of dimensions while preserving the most important information. By projecting the data onto a smaller set of principal components, PCA can make high-dimensional data more manageable and interpretable, improve the performance of machine learning algorithms, and reduce the risk of overfitting.

This paper aims to provide a practical investigation into the mechanics and applications of PCA. Through hands-on exercises, we will explore how PCA works, how to implement it using popular programming tools, and how to interpret the results.

II. RELATED WORK

Dimensionality reduction techniques play a crucial role in machine learning, with Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) being prominent approaches for reducing dimensionality and visualizing data.

PCA stands as a cornerstone of this field, offering a powerful and interpretable approach [1]. Introduced by Karl Pearson in 1901 [2], PCA aims to identify a new set of orthogonal directions, known as principal components (PCs), that capture the greatest variance in the data [2]. These PCs represent the directions of the greatest information content, allowing for dimensionality reduction while preserving the most critical information in the data [1]. PCA's effectiveness in data compression and feature extraction has led to its widespread adoption in diverse fields, including signal processing [3], image compression [4], and machine learning [5].

The mathematical foundation of PCA lies in linear algebra, particularly in singular value decomposition (SVD) [6]. SVD allows for efficient computation of the principal components and their associated variances (eigenvalues) [6]. The seminal work by Wold et al. (1987) provides a detailed explanation of PCA's mathematical underpinnings and its application in chemometrics [7]. Beyond its core functionality, PCA offers valuable insights into data structure. By analyzing the explained variance captured by each principal component, researchers can gain an understanding of the underlying factors driving the data's variability [1]. Additionally, PCA can be used for outlier detection, as data points deviating significantly from the principal components may represent anomalies or errors [8].

For tasks where preserving local structure is crucial, non-linear dimensionality reduction techniques become valuable. t-SNE is a popular technique specifically designed for the visualization of high-dimensional data [9]. Unlike PCA, t-SNE focuses on preserving the similarities between data points in the high-dimensional space by mapping them onto a lower-dimensional space while maintaining these relationships [9]. This property makes t-SNE particularly useful for visualizing complex datasets and identifying

clusters or patterns that may be obscured in the original data [9].

UMAP is a relatively recent advancement in dimensionality reduction that combines elements of both linear and non-linear methods [10]. Similar to t-SNE, UMAP aims to preserve both local and global structures in the data while creating a low-dimensional representation [10]. It achieves this by constructing a high-dimensional graph representation of the data and then optimizing a low-dimensional graph to closely approximate the structure of the original data [10]. UMAP has gained significant attention for its ability to efficiently capture complex relationships in high-dimensional datasets, making it a valuable tool for tasks like clustering, visualization, and manifold learning [10].

These techniques offer distinct advantages for dimensionality reduction. PCA provides a powerful linear approach for capturing variance, while t-SNE and UMAP excel at preserving local and potentially global structures in complex, non-linear datasets. Choosing the most suitable technique depends on the specific characteristics of the data and the desired outcome of the analysis.

III. METHODOLOGY

A. DATASET DESCRIPTION

The comprehension, analysis and evaluation of dimensionality reduction using Principal Component Analysis (PCA) were performed using synthetic dataset and then using a standard dataset from *UC Irvine Machine Learning Repository*.

The first dataset was an artificially synthesized 20x2 array dataset generated using NumPy's 'randn' function that produced random values sampled from a standard normal distribution. Consequently, the data was centered around a zero mean and unit standard deviation. This dataset was initially used to introduce the fundamentals of PCA and for easy visualization in a controlled environment.

The second dataset that we used was an educational administration dataset named '*Predict Students' Dropout and Academic Success*' [11]. The dataset is represented using a tabular structure and falls within the subject area of social science. It is obtained from multiple separate databases of a higher education institution and pertains to students enrolled in various undergraduate programs such as agronomy, design, education, nursing, journalism, management, social service, and technology [11]. It encompasses information available at the time of enrollment (including academic history, demographics, and socio-economic factors) along with students' academic performance at the end of their first and second semesters [11].

The dataset consists of 36 features and 4414 instances. Each instance represents a student. The dataset underwent rigorous data preprocessing to handle data from anomalies, unexplainable outliers, and missing values [11]. There are no missing values in the dataset.

The dataset was created in a project that aims to contribute to the reduction of academic dropout and failure in higher education, by using machine learning techniques to identify students at risk at an early stage of their academic path, so that strategies to support them can be put into place [11]. The dataset includes information known at the time of student enrollment – academic path, demographics, and social-economic factors.

The problem is formulated as a three-category classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course, characterized by a significant imbalance toward one of the classes [11].

B. PROPOSED METHODOLOGY

Principal Component Analysis (PCA) is a statistical technique that transforms a dataset's features into a set of uncorrelated variables known as principal components, where each component captures a portion of the dataset's variance in descending order.

In our study, PCA is applied to the dataset to understand and visualize its structure. The process begins with encoding the target labels into numerical values using a mapping dictionary. The dataset is then converted into a matrix form, while also dropping the target feature, enabling matrix operations after which, a pair plot is generated to visualize relationships between attributes, providing initial insights into the data structure.

The dataset then undergoes standardization by subtracting the mean of each attribute with each attribute value to ensure that the data has zero mean and unit variance. The covariance matrix of the standardized data is calculated to understand the relationships between different attributes. The covariance matrix is also used to compute Eigenvalues and eigenvectors. The eigenvalues and eigenvectors, sorted in descending order of the eigenvalues, helps to identify the principal components that capture the most variance.

The standardized dataset is multiplied with sorted eigenvectors to achieve a change of biases which transforms the data into new principal component space. PCA is then applied to project the transformed data onto the principal components.

Finally, scatter plots and 3D visualizations of the projected data are generated to visualize the data in the new principal component space.

This approach helps to thoroughly analyze and visualize the dataset using PCA. It shows the main patterns and connections within the data. We employed Principal Component Analysis (PCA) as the primary method for dimensionality reduction.

The steps involved in the PCA algorithm are outlined below:

Given centered training data $X \in \mathbb{R}^{N \times P}$ and a desired lower dimension K .

Calculate the covariance matrix $C = X^T X$.

Calculate the first K largest eigenvalues of C and their corresponding eigenvectors $u_1, u_2, \dots, u_K$.

Build the PCA basis $U \in \mathbb{R}^{P \times K}$, $U = [u_1, u_2, \dots, u_K]$.

To transform a new sample x^* :

Return $z^* = U^T x^*$

C. MATHEMATICAL FORMULAE

Let's assume we have a matrix or dataset with dimensions $m \times n$.

The formula for the mean (μ) is:

$$\mu = \frac{\sum x_{ij}}{m \times n} \quad (1)$$

Where x_{ij} represents the element at the i -th row and j -th column of the matrix.

The formula for the covariance matrix Σ is:

$$\Sigma = \frac{1}{m-1} \cdot (X^T \cdot X) \quad (2)$$

For a square matrix A , the eigenvalues can be obtained by solving the characteristic equation:

$$|A - \lambda I| = 0 \quad (3)$$

Where λ is the eigenvalue and I is the identity matrix of the same size as A .

Given its eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, the eigenvectors can be found by solving the equation:

$$(A - \lambda_i I)v_i = 0 \quad (4)$$

The proportion of variance (PoV) for each eigenvalue λ_i is given by:

$$PoV_i = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j} \quad (5)$$

Where $\sum_{j=1}^n \lambda_j$ represents the sum of all eigenvalues.

D. INSTRUMENTATION DETAILS

The analysis and visualization in this study were conducted using *Jupyter Notebook*, a versatile and interactive web-based coding environment. Numeric operations and dataset manipulations were performed using the *NumPy* library, while the *Pandas* library was utilized for importing the dataset and conducting Exploratory Data Analysis (EDA). For data visualization, *Matplotlib* was employed to plot data points on two-dimensional principal component axes, and its submodule '*mpl_toolkits.mplot3d*' was used for visualizing the data points in three-dimensional principal component axes. In summary, *Jupyter Notebook*, along with these powerful libraries, facilitated efficient and effective experimentation.

E. SYSTEM BLOCK DIAGRAM

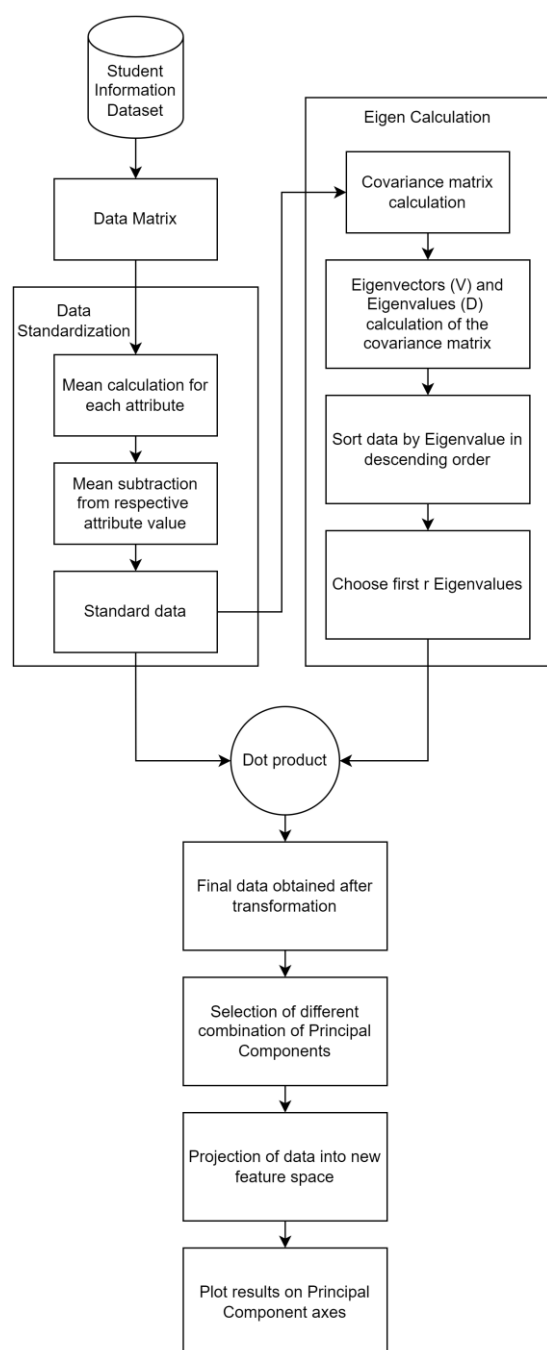


Figure 1: Block Diagram of Principal Component Analysis

IV. EXPERIMENTAL RESULTS

A. PCA ON RANDOM DATA SAMPLED FROM STANDARD NORMAL DISTRIBUTION

Principal Component Analysis (PCA) was conducted on synthesized data to demonstrate the effectiveness of PCA in reducing the number of dimensions by capturing the maximum variance with minimal loss of information. Several 2D plots were generated to visualize the different stages of the PCA analysis.

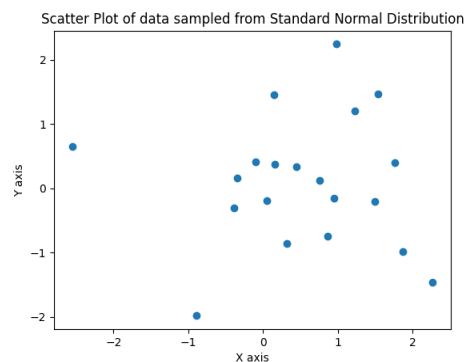


Figure 2: Plot of Synthesized Data

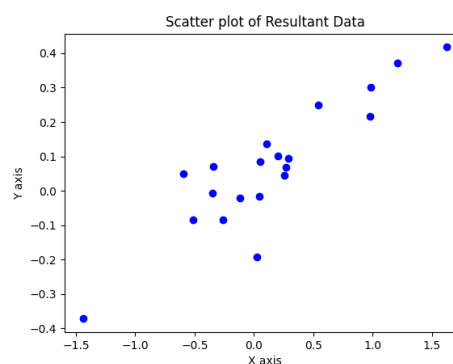


Figure 3: Plot of Resultant Data

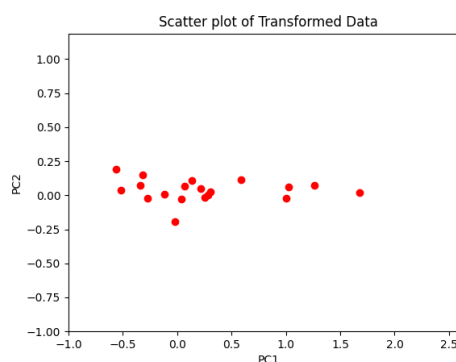


Figure 4: Plot of Data Transformed using PC1 and PC2

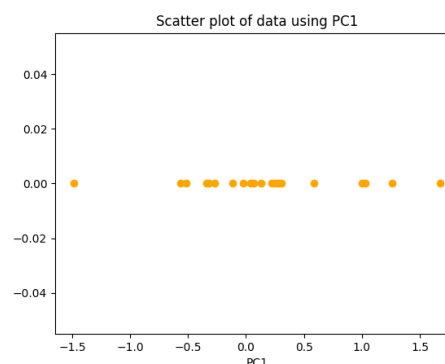


Figure 5: Plot of Data Transformed using PC1 only

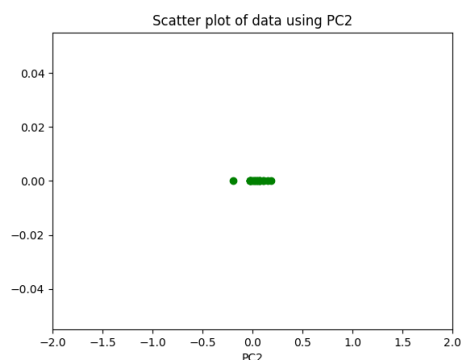


Figure 6: Plot of Data Transformed using PC2 only

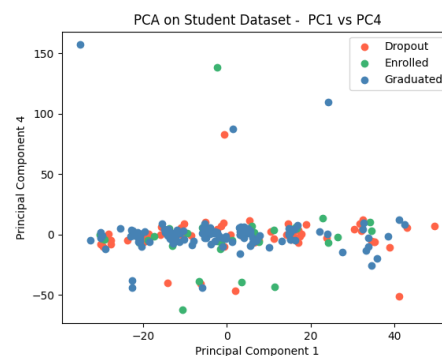


Figure 9: Plot using PC1 and PC4

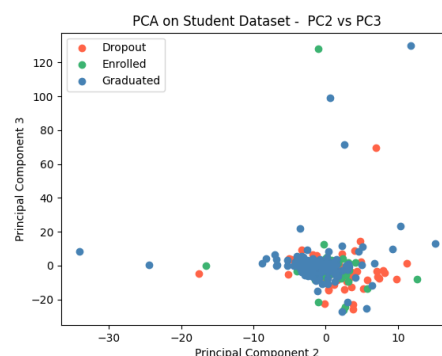


Figure 10: Plot using PC2 and PC3

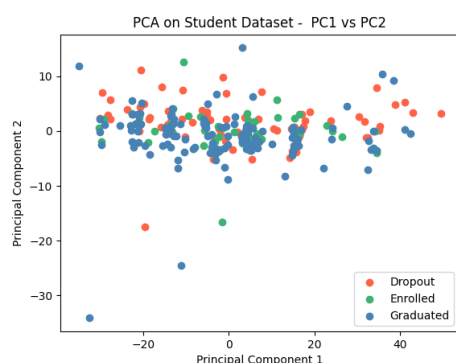


Figure 7: Plot using PC1 and PC2

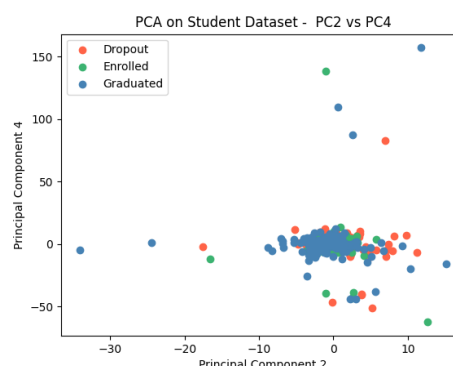


Figure 11: Plot using PC2 and PC4

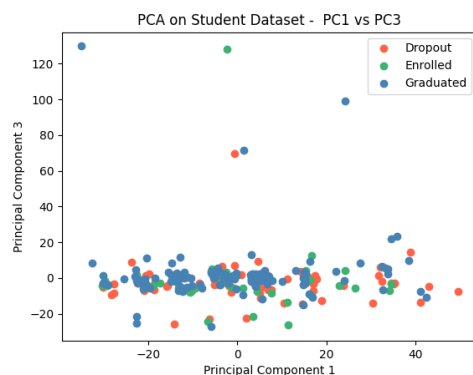


Figure 8: Plot using PC1 and PC3

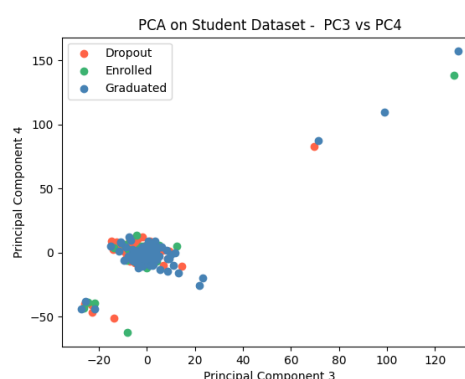


Figure 12: Plot using PC3 and PC4

B. PCA ON 'PREDICT STUDENTS' DROPOUT AND ACADEMIC SUCCESS' DATASET

Principal Component Analysis (PCA) was conducted on a real-world dataset to demonstrate how complex data can be simplified and effectively visualized, highlighting the ability of PCA to reduce dimensionality while preserving essential patterns and relationships within the data. Different plots were created to understand the spread of data across various dimensions and to observe how the principal components capture the underlying structure and variance of the dataset.

1. 2D PLOTS OF STUDENT DATASET

2. SCREE PLOT FOR STUDENT DATASET

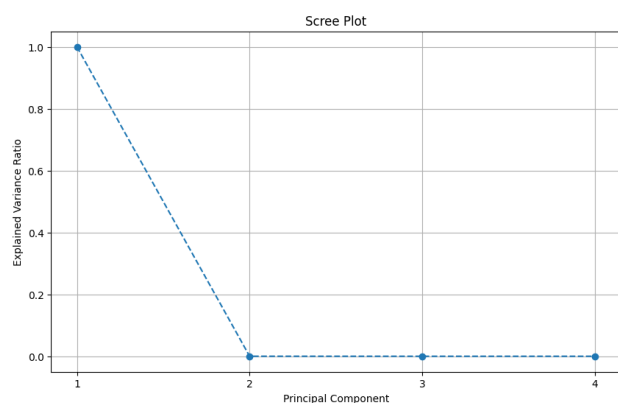


Figure 13: Scree Plot of Principal Components

3. 1D PLOT OF STUDENT DATASET

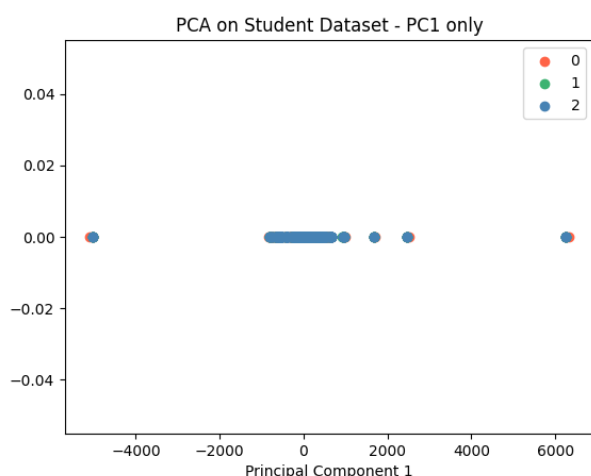


Figure 14: Plot using PC1 only

4. 3D PLOTS OF STUDENT DATASET

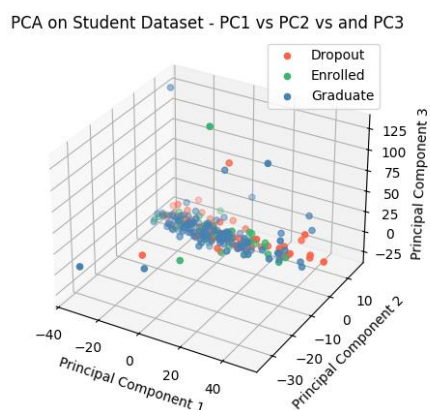


Figure 15: 3D Plot using PC1, PC2 and PC3

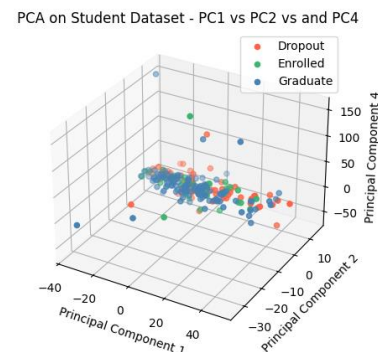


Figure 16: 3D Plot using PC1, PC2 and PC4

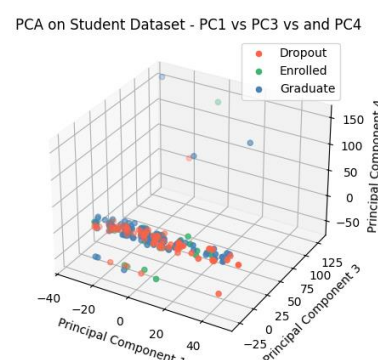


Figure 17: 3D Plot using PC1, PC3 and PC4

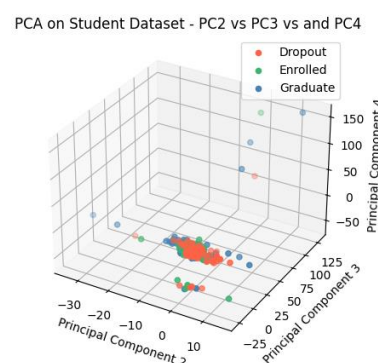


Figure 18: 3D Plot using PC2, PC3 and PC4

V. DISCUSSION AND ANALYSIS

A. PCA ON RANDOM DATA SAMPLED FROM STANDARD NORMAL DISTRIBUTION

In IV A, we synthesized a 20x2 dataset sampled from a standard normal distribution using Python, with a random seed set to '0'. Initially, the data appeared randomly scattered with no discernible structure, as depicted in Figure 2. To introduce structure, we transformed the data by multiplying it with a 2x2 matrix whose values were sampled from a uniform distribution. The resulting dataset aligned along an axis, as shown in Figure 3.

We calculated the variance along the data axes and observed that the variance along the X-axis was significantly greater than that along the Y-axis. Next, we computed the covariance matrix, but it did not exhibit the desired properties: the diagonal elements were not maximized, and the off-diagonal elements were not minimized.

To address this, we computed the eigenvalues and eigenvectors of the covariance matrix and sorted the eigenvectors in descending order based on their corresponding eigenvalues. Initially, we retained both eigenvectors and performed a change of basis by multiplying the data with the eigenvector matrix. This transformation resulted in a covariance matrix with the desired properties: the diagonal elements were maximized, and the off-diagonal elements were minimized. We also calculated the proportion of variance explained by each eigenvector, discovering that the first principal component accounted for approximately 98.75% of the total variance, while the second principal component accounted for approximately 1.25%. This distribution of variance is illustrated in Figure 4, where the spread of the data is maximal along the first principal component axis compared to the second.

Subsequently, we performed a change of basis using only one eigenvector at a time. The 1D plots of the transformed data using the first principal component and the second principal component are shown in Figures 5 and 6, respectively. As evident, the spread of data along the first principal component is maximal, highlighting its significance in capturing the dominant variation in the dataset. This underscores the importance of the first principal component in effectively summarizing the structure and variance of the data.

B. PCA ON 'PREDICT STUDENTS' DROPOUT AND ACADEMIC SUCCESS' DATASET

In IV B, we conducted dimensionality reduction on a standard dataset titled '*Predict Students' Dropout and Academic Success*'. The dataset initially featured numerous attributes, posing challenges in discerning underlying patterns and structures. To address this complexity and facilitate better analysis, we employed Principal Component Analysis (PCA), which helped to extract meaningful patterns while minimizing the loss of critical information.

Before performing PCA, we performed some preprocessing. The data was already rigorously preprocessed by the authors of the dataset. So, there was no need to handle missing values, outliers, and redundancy. We split the data into x and y , with x containing the features and y containing the target feature. All the features in x were numerical, so no encoding was needed. The target feature was categorical. So, we performed label encoding by mapping the three unique target variables to numbers 0, 1, and 2. Finally, PCA was performed.

The data x was standardized to achieve a zero mean. Upon computing the covariance matrix, we observed that it lacked desirable properties. To address this, we computed the eigenvectors matrix, sorted it in descending order based on eigenvalues, and projected the standardized data onto the eigenvectors. This transformation yielded a covariance matrix with desirable properties, indicating successful PCA application.

First, we visualized the data in 2D space using various combinations of the first four principal components (PC1 to PC4), as depicted in Figure 7 to Figure 12. Notably, plots in Figure 7, 8, and 9 exhibited maximum data spread along one principal component axis, predominantly involving PC1. Conversely, plots in Figure 10, 11, and 12 showcased minimum data spread along one principal component axis. A common pattern was discovered: PC1's pivotal role in maximizing data spread.

To further investigate this, we visualized the proportion of variance by each principal component using a scree plot as shown in Figure 13. From the scree plot, we discovered that the first principal component PC1 accounted for approximately 100% of the total variance in the data. Motivated by this finding, we visualized the data in 1D space using PC1 exclusively, as illustrated in Figure 14.

Furthermore, to explore interrelationships among multiple attributes, we employed 3D plots (Figure 15 to 18), aiming to elucidate complex data interactions.

VI. CONCLUSION

In this paper, we explored Principal Component Analysis (PCA) as a foundational technique for dimensionality reduction. Initially, we applied PCA to random data to understand its basic principles and mathematical underpinnings. Through this exploration, we gained insights into how PCA identifies orthogonal directions capturing the highest variance in the data, enabling dimensionality reduction while preserving critical information.

Following the exploration with random data, we applied PCA to a standard dataset to further illustrate its practical utility. By analyzing the explained variance captured by each principal component, we gained a deeper understanding of the underlying factors driving the data's variability. Additionally, PCA facilitated data visualization, feature extraction, and outlier detection in the standard dataset, highlighting its effectiveness in real-world scenarios.

While PCA served as the primary focus of this lab report, it's important to acknowledge the existence of other dimensionality reduction techniques, such as t-Distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP). These techniques offer distinct advantages, particularly in preserving local and non-linear structures in high-dimensional data, which could be explored in future analyses. By gaining proficiency in PCA and other dimensionality reduction techniques, researchers can

effectively manage high-dimensional datasets, extract meaningful insights, and improve the performance of machine learning models. This lab report provides a solid foundation for understanding and applying PCA, paving the way for further exploration and experimentation in the field of dimensionality reduction.

ACKNOWLEDGEMENT

We extend our heartfelt appreciation to the Department of Electronics and Computer Engineering at IOE, Thapathali Campus for providing us with the invaluable opportunity to engage in learning, research, and the realization of our insights. We would like to express our sincere gratitude to our esteemed lecturer Er. Dinesh Baniya Kshatri for providing unwavering support, invaluable guidance, and insightful suggestions throughout our academic career.

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BIOGRAPHY



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